MILITARY SPECIFICATION

INDICATOR SET, LIQUID OXYGEN QUANTITY, A/A24J-21

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of 5-liter liquid oxygen quantity indicator set that is designated A/A24J-21. The indicator set consists of a master indicator and a repeater indicator.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-25645	Indicator, Liquid Oxygen Quantity, Capacitance Type, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-831	Test Reports, Preparation of
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:13Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General specification. Unless otherwise specified herein, the requirements of section 3 of MIL-I-25645 shall apply as requirements for this specification. If the requirements of section 3 of MIL-I-25645 and the requirements of section 3 of this specification conflict, the requirements of this specification shall govern.

3.1.1 Qualification. The indicator sets furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.1.2 and 6.3).

3.1.2 Components. The indicator set shall consist of a master indicator and a repeater indicator. The master indicator and the repeater indicator shall include all of the equipment (except cables) necessary to meet the performance requirements specified herein and in MIL-I-25645.

3.1.3 Capacitance. For design purposes, the capacitance for zero reading shall be 63.5 picofarads (pF).

3.1.4 General electrical requirements

3.1.4.1 Power-failure flag. The master indicator and the repeater indicator shall each have a power-failure flag. The flags shall be blank for power-on conditions and shall indicate "OFF" for power-off conditions.

3.1.4.2 Integral lighting. The master indicator and the repeater indicator shall each have integral lighting. The integral lighting shall conform to MIL-L-27160.

3.1.4.3 Electrical connectors. The electrical connectors shall be in accordance with figures 1 and 2. The master indicator and the repeater indicator shall be designed internally so that they will function properly when the mating connectors are wired as shown in figures 1 and 2. The part numbers of the connectors that are specified in figures 1 and 2 are designations of the Amphenol Connector Division; 2801 South 25th Avenue; Broadview, Illinois 60153.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:13Z
Check the source to verify that this is the current version before use.

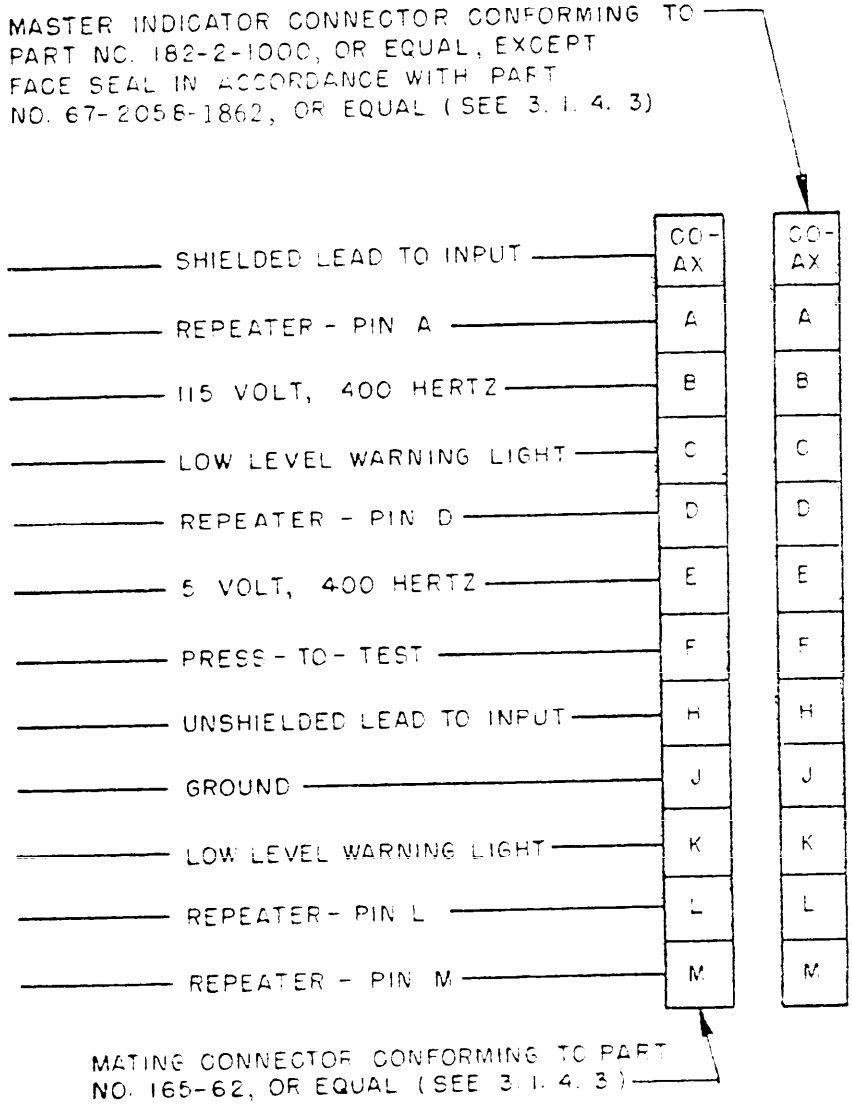


FIGURE 1. Master Indicator Connector Wiring

Source: https://assist.dla.mil -- Downloaded: 20260-23T08:13Z
Check the source to verify that this is the current version before use.

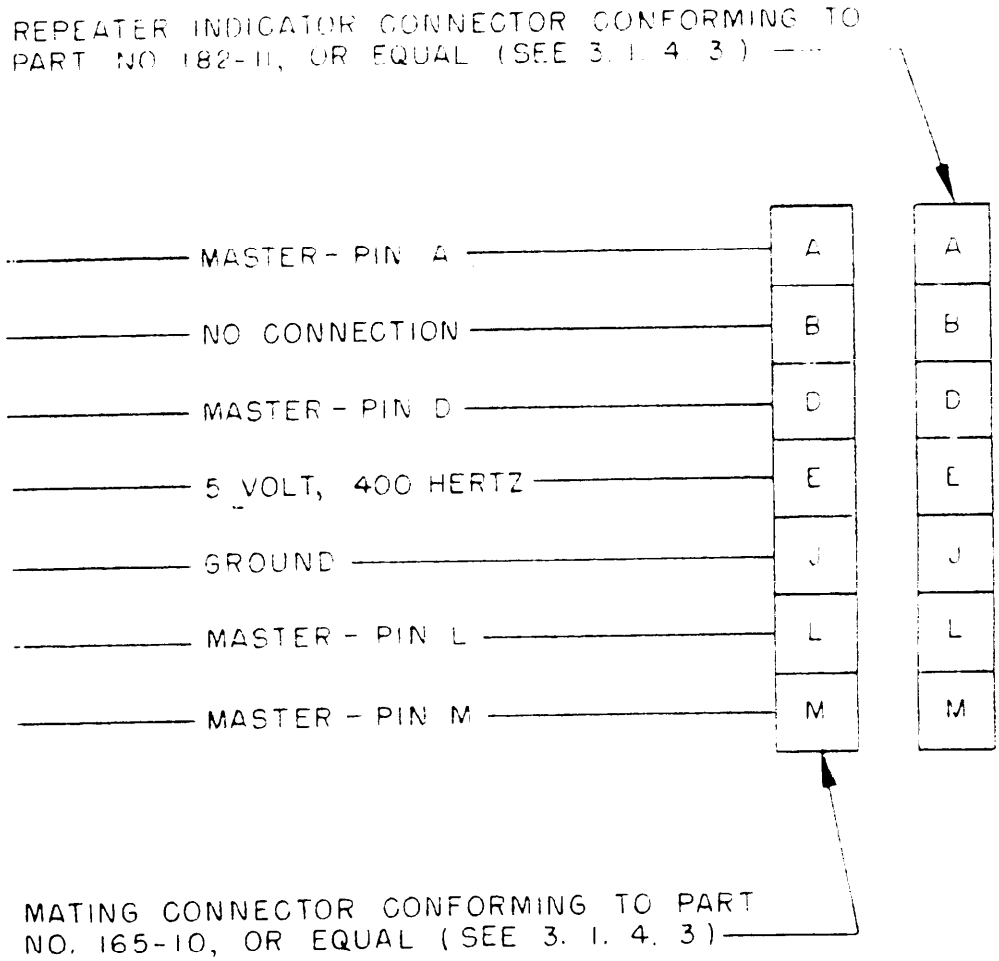


FIGURE 2. Repeater Indicator Connector Wiring

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:13Z
Check the source to verify that this is the current version before use.

3.1.5 Performance

3.1.5.1 Accuracy. When the indicator set is tested as specified in MIL-I-25645, the accuracy of the master indicator shall be in accordance with table I. The tolerance for the room-temperature reading is the allowable deviation from the nominal calculated values. The tolerance for the repeater indicator is the allowable deviation from the corresponding master indicator reading. Any other tolerances are the allowable deviation from the room-temperature reading.

TABLE I. Tolerances for Capacitance Readings

Performance Requirements	Tolerance (Plus or Minus)
Accuracy - room temperature	0.4 pF
Low temperature operation	0.5 pF
High temperature operation	0.5 pF
Vibration	0.3 pF
Power supply variation	0.3 pF
Low temperature storage and operation	0.5 pF
High temperature storage and operation	0.5 pF
Salt spray	0.3 pF
Cycling	0.3 pF
Repeater	0.3 pF

3.1.5.2 Calibration adjustments. The zero calibration adjustment for both the master indicator and the repeater indicator shall be capable of adjusting the pointer to zero indication over an input capacitance range (to the master indicator) of at least 60.0 pF to 67.0 pF. With the indicator adjusted to zero indication at 60.0 pF input capacitance, the indicator shall be capable of calibration to full scale indication at 89.0 pF input capacitance by means of the full calibration adjustment. With the indicator adjusted to zero indication at 67.0 pF input capacitance, the indicator shall be capable of calibration to full scale indication at 96.0 pF input capacitance by means of the full calibration adjustment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:13Z
Check the source to verify that this is the current version before use.

3.1.5.3 Power supply variation. When the indicator is tested as specified in MIL-I-25645, variations of supply voltage from 105 volts to 125 volts and frequency of 320 to 480 hertz shall not cause indicator errors exceeding those specified in table I. For the 105-volt test point, the indicator shall be checked at frequencies of 320 and 480 hertz. For the 125 volt test point, the indicator shall be checked at frequencies of 320 and 480 hertz.

3.1.6 Details of components

3.1.6.1 Cases. The case of the master indicator shall not exceed 5.0 inches in length. The case of the repeater indicator shall not exceed 3.5 inches in length.

3.1.6.2 Dials. The dial of the master indicator and the dial of the repeater indicator shall conform to figure 3. The numerals and the letters on the dials shall conform to MS33558.

3.1.7 Weight. The weight of the master indicator shall not exceed 1.5 pounds. The weight of the repeater indicator shall not exceed 0.75 pound.

4. QUALITY ASSURANCE PROVISIONS

4.1 General specification. Unless otherwise specified herein, the quality assurance provisions for the indicator set shall be in accordance with section 4 of MIL-I-25645. In the event of conflict between the quality assurance provisions of MIL-I-25645 and the quality assurance provisions of this specification, the quality assurance provisions of this specification shall govern.

4.1.1 Classification of tests. The indicator set shall be subjected to the following:

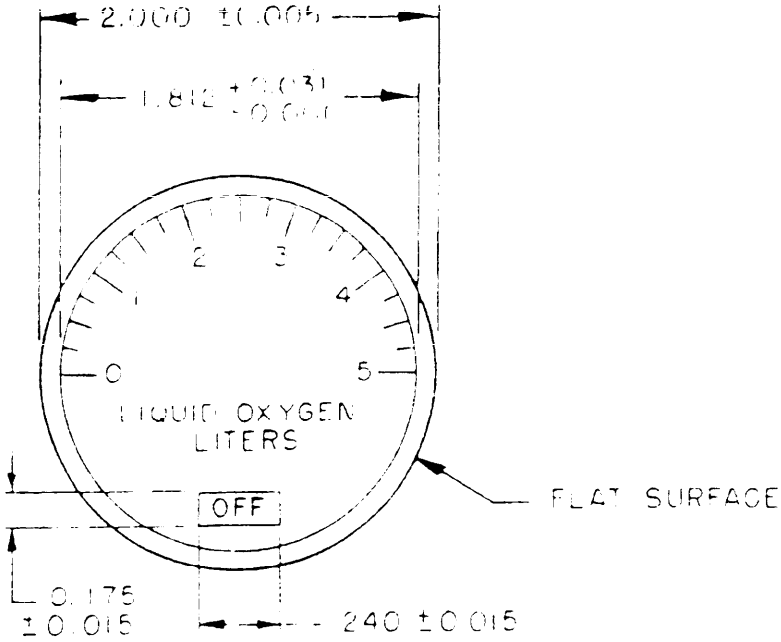
(a) Qualification tests

(b) Quality conformance tests.

4.1.2 Qualification tests. One indicator set shall be subjected to the qualification tests which shall consist of all the examinations and tests described under "Test Methods" of MIL-I-25645.

4.1.2.1 Qualification sample and test report. The indicator set that has been tested as specified in 4.1.2 and three copies of a qualification test report that has been prepared in accordance with MIL-STD-831 shall be submitted to the qualifying activity (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 202640-23T08:13Z
Check the source to verify that this is the currentversion before use.



MARKINGS ON DIAL	HEIGHT OR LENGTH (±0.010)	WIDTH OF LINE OR GRADUATION (±0.005)	COLOR OF FINISH (SEE FED-STD-595)
NUMERALS	0.140	0.025	37875
GRADUATIONS			
MAJOR	0.188	0.031	37875
MINOR	0.125	0.015	37875
SUB MINOR	0.080	0.015	37875
LETTERS			
LIQUID OXYGEN	0.140	0.020	37875
LITERS	0.100	0.015	37875
OFF	0.062	0.015	37875
	0.094	0.015	37875
BACKGROUND			
DIAL			37038
OFF FLAG			31158

DIMENSIONS IN INCHES

FIGURE 3. Dial

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:13Z
Check the source to verify that this is the current version before use.

4.1.2.2 Qualification reliability testing. The indicator set shall be tested for reliability and shall be continued through a longevity test in accordance with MIL-STD-781. Acceptance shall be judged on the basis of the accept-reject criteria of test plan I of MIL-STD-781. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation for delivery of the indicator set shall be in accordance with section 5 of MIL-I-25645. The levels of preservation and packaging and packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The indicator set covered by this specification is intended to indicate the quantity of liquid oxygen remaining in the converters of certain F-5F aircraft.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Selection of applicable levels of preservation and packaging and packing (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:13Z
Check the source to verify that this is the current version before use.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturer's are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division (ASD/ENCCE), Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

International Interest
(see MIL-I-25645)

Preparing activity:
Air Force - 11

Project No. 6680-F161

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:13Z
Check the source to verify that this is the current version before use.